

Seungwon Jeong

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Employment

- **Postdoctoral Research Fellow with Jaegi Jeon**
Global-Learning & Academic research institution for Master's · PhD students, and Postdocs, Chonnam National University, Republic of Korea, Sep. 2024 - present.
- **Postdoctoral Research Fellow with Jeonggyu Huh**
Department of Mathematics, Sungkyunkwan University, Republic of Korea, Mar. 2024 - Aug. 2024
- **Postdoctoral Research Fellow with Minsuk Kwak**
Department of Mathematics, Hankuk University of Foreign Studies, Republic of Korea, Sep. 2023 - Feb. 2024

Education

- **Ph.D. with Hyeng Keun Koo**
Department of Financial Engineering, Ajou University, Republic of Korea, Mar. 2018 - Aug. 2023
- **Bachelor's degree in Mathematics and Business Administration**
Department of Mathematics, Chonnam National University, Republic of Korea, Mar. 2011 - Feb. 2018

Research Interest

- Empirical asset pricing, Portfolio optimization, Machine learning, Deep learning, Generative AI, Reinforcement learning, Big Data Analysis

Publications

- Equity Premium Forecasting with Reliability-Screened Forward-Looking Signals (with Jeonggyu Huh* and Jaegi Jeon), *PLOS One*, 21(5), e0341578 (2026)
- DeepONet-based Surrogate Modeling for Bond Option Pricing (with Sanghyun Lee* and Jeonggyu Huh), *AIMS Mathematics*, 11(3), 5853-5896 (2026)
- Optimal Investment under Irreversible Consumption and Locally Risk-Seeking Preferences (with Junkee Jeon* and Hyeng Keun Koo), *Journal of Industrial and Management Optimization*, 22(2), 1063-1086 (2026)
- LSTM-based Dynamic Correlation Forecasting with Economic Conditions (with Seungwoo Ha, Jeonggyu Huh*), *Finance Research Letters*, 108425 (2025)
- Pontryagin-Guided Direct Policy Optimization for Continuous-time Portfolio Problem (with Jeonggyu Huh* and Jaegi Jeon*), *Journal of Industrial and Management Optimization*, 0-0 (2025)
- Learning Distributions for Continuous-Time Financial Models (with Jeonggyu Huh*), *Computational Economics*, 1-35 (2025)

- Dual-Uncertainty Modeling in Financial Time-Series via VMD-LSTM with Concrete Dropout and VMD-WGAN (with Jeonggyu Huh*, Dajin Kim, and Minseok Jung), *Networks and Heterogeneous Media*, 20(5), 1411-1436 (2025)
- Accelerating SDE Simulation through Learning of Stochastic Dynamics (with Ji-Hun Kim, Jitae Jung, and Jeonggyu Huh*), *Journal of Korean Society for Industrial and Applied Mathematics* (2024)
- Asset allocation strategy and convergence with Epstein-Zin utility function (with Seryoong Ahn, Sang Jin Ahn, Hyeng Keun Koo*) , *Korean Management Science Review*, 39(3), 1-14 (2022).
- Optimal portfolio choice in a Binomial-tree and its convergence (with Seryoong Ahn*, Sang Jin Ahn, Hyeng Keun Koo), *East Asian Mathematical Journal*, 38(3), 277-292 (2022).

Submitted papers

- MarketGANs: Multivariate Financial Time-Series Data Augmentation Using Generative Adversarial Networks, under revision at *Financial Innovation* (with Jeonggyu Huh, Hyeng-Keun Koo, Byung-Hwa Lim, and Hyun-Gyoon Kim)
- Neural Policy Iteration for Dynamic Portfolio Choice with Control-Dependent Diffusion (with Jeonggyu Huh and Yeoneung Kim)
- Self-Consistent Adjoint Policy Iteration for Constrained Dynamic Portfolio Choice (with Jeonggyu Huh and Yeoneung Kim)
- Scalable Dynamic Portfolio Allocation via Physics-Informed Neural Networks (with Yeoneung Kim and Jeonggyu Huh)

Working papers

- Pontryagin-Guided Policy Iteration for Discrete-Time Infinite-Horizon Portfolio Choice (with Jeonggyu Huh)
- Structure-Preserving Policy Iteration for Dynamic Portfolio Choice: Whole-Space Convergence, Localization, and Neural Certification (with Jeonggyu Huh and Yeoneung Kim)
- Recovering the Kinks in Delay Control: A Structure-Aware Optimal Control Solver with Pontryagin Projection (with Jeonggyu Huh and Yeoneung Kim)
- Adversarial Time-Series Domain Adaptation for Early-Stage IPO Price Prediction (with Youngwoo Lee and Jeonggyu Huh)
- Investment with Taxes (with Hyeng Keun Koo, Jeonggyu Huh, and Byung Hwa Lim)

Projects

- Developing an Extended Economic Scenario Generator with Machine Learning, Korea Asset Pricing (KAP), Aug. 2024 - Jan. 2025

Talks

- KSIAM 2026 Spring Conference, May 2026, UNIST, Ulsan, Republic of Korea.
- KMS 2025 Spring Meeting, Apr. 2025, KAIST, Daejeon, Republic of Korea.

- KSIAM 2024 Spring Conference, May 2024, Global Plaza, KyungPook National University, Daegu, Republic of Korea.
- KSIAM 2023 Annual Meeting, Nov. 2023, Kimdaejung convention center, Gwangju, Republic of Korea.
- KSIAM 2023 Spring Conference, May 2023, Alpensia, Pyeong Chang, Republic of Korea.
- KSIAM 2022 Annual Meeting, Nov. 2022, Sono Calm, Jeju, Republic of Korea.
- KSIAM 2021 Annual Meeting, Dec. 2021, Bexco, Busan, Republic of Korea.
- KAFE-SKKU Winter Conference of Finance (Online), Dec. 2021, Republic of Korea.

Language and Skills

- Programming skills: Python, SAS, R, MATLAB